

For this week's assignment, I tried to build a self-playing battle game that mixes physics chaos with a simple player prediction mechanic. Instead of designing traditional fighters, I thought it would be more fun and honestly more playable to abstract the idea of combat into bouncing colored balls on an irregular platform. The game basically turns into a physics-based survival match where players watch ten different balls bumping, bounce, and eventually fall off the stage until only one remains. I liked the concept because it breaks away from expected "battle" tropes and honestly it is easier to build the whole system. Adding player-controlled fans on the left and right sides gave the game just enough interaction to keep it entertaining and immersive without breaking the "self-playing" requirement.

While building the project, I ran into way more problems than I originally expected. My first idea involved cars with rigidbodies crashing into each other, but that turned into a disaster for me, because cars flipped, jittered, got stuck, or fly into the sky. I also pulled a unity package for cars' auto drive system in unity assets store, but due to the version of this is too old, it falls. Switching to spheres instantly solved most physics issues, because spheres will keep rolling and never get stuck and behave more predictably. Another challenge was keeping the balls always come to the mid and bump together. I solved this by giving every ball a small script that constantly applies randomized forces, plus a center-biased force to keep them from drifting away..

Next week, I think I need to be more disciplined about scoping. I wasted hours trying to make my car auto bumping ideas that simply wasn't worth saving. Also, based on the feedbacks I received today, I will make each ball have their own materials to make sure none of them have similar color. Further more, I'll decrease the interval of check how many balls still on the ground to make sure only 1 ball wins.

As for the future potential of this project, I think the game could actually become a small party-style game. I can imagine adding different ball types, power-ups, multiple lanes, or even "abilities" that trigger automatically. But at the same time, I'm also aware that this game's traits comes from its simplicity and chaos. If I add too much, the identity might get lost. For me, I think I learned a lot on this game style. I used to think these self-combat game is boring, but I know there are a lot ways to make it interesting.